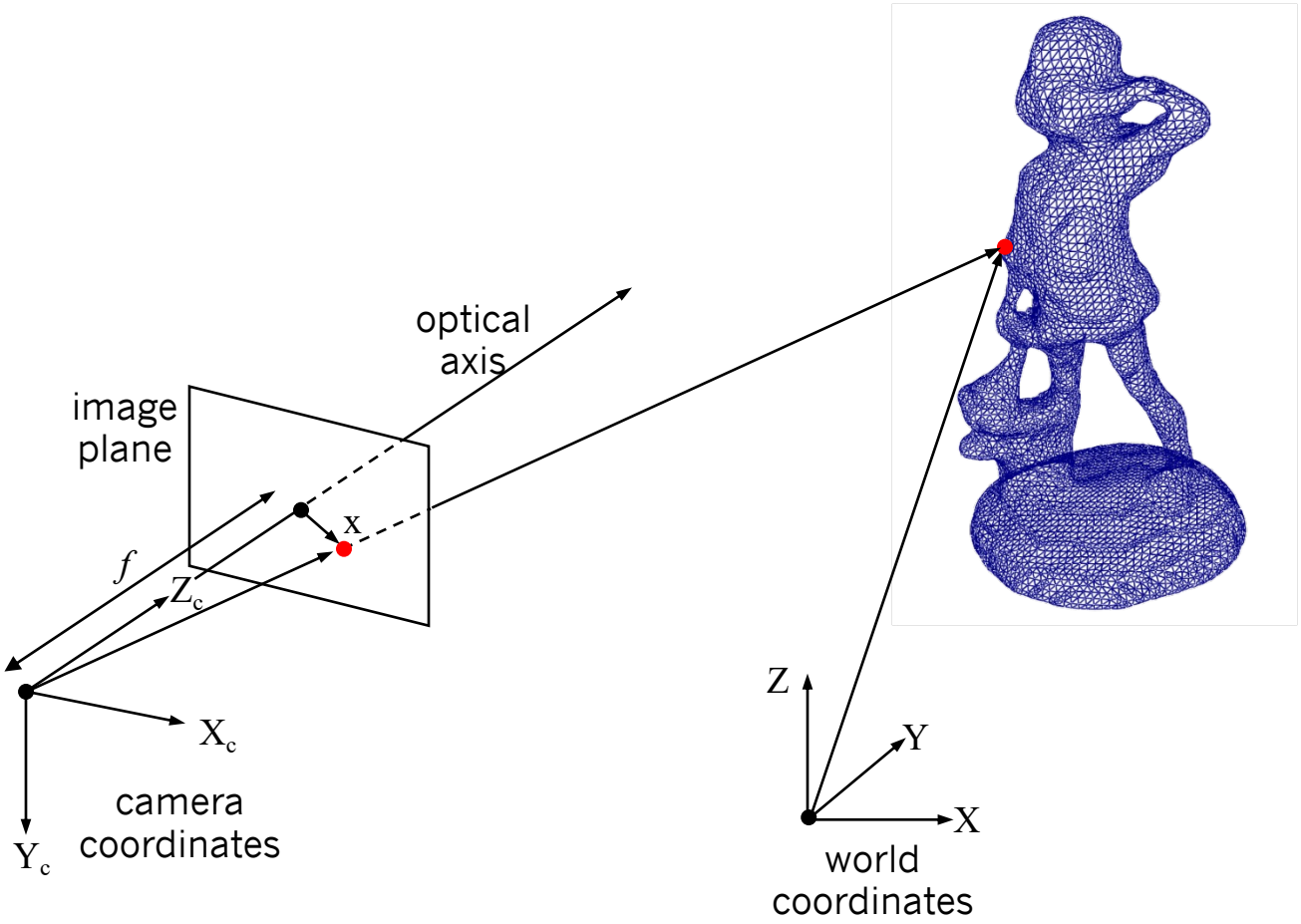
Course Information

COMP3317 Computer Vision

Dr. Kenneth Wong

Computer Vision



Who is who?

- Instructor and Tutor

	Instructor	Tutor	Tutor
Name:	Kenneth Wong	Hao Shaozhe	Lyu Zhengyao
Office:	CB323	CB411	CB411
Tel:	2857 8443	2857 8450	2857 8450
Office hours:	Thu, 14:00 – 16:00	Mon, 14:00 - 16:00	Tue, 14:00 - 16:00

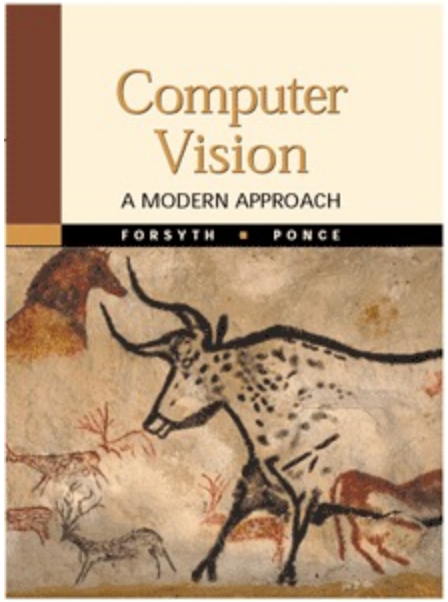
Course schedule

- Lectures and Tutorials

Lectures	Thu	16:30 – 18:20	LE1
Tutorials	Mon	17:30 – 18:20	LE1

Resources

- Reference book
  - Computer Vision – A Modern Approach
    - by Forsyth & Ponce
- Lecture Notes
  - Available on the course Moodle page
- Tutorial Notes
  - Available on the course Moodle page



What do we need to know before taking this course?

- Prerequisites
  - Basic vector and matrix operations
    - Addition, subtraction, multiplication of matrices and vectors
    - Dot product and cross product of vectors
    - Inverses and transposes of matrices
    - Eigenvalues and Eigenvectors
  - Solving systems of linear equations
  - Basic calculus (differentiation)
  - Python programming

What are we going to learn?

- Topics to be covered
  - Digital image processing
  - Feature extraction
  - Camera models
  - Image invariants
  - Camera calibration
  - Stereo vision
  - Image classification using deep learning

What are the outcomes?

- Course Learning Outcomes

Image processing	Students understand how images are digitally represented and learn how to perform image processing.
Feature extraction	Students are able to implement low-level edge and corner detection algorithms.
Camera model and calibration	Students are able to model a pinhole camera and discover the intrinsic and extrinsic camera parameters.
Stereo vision	Students are able to extract 3D information from images and understand fundamental concepts in multiple view geometry.

Assessments

- 3 written assignments (10% each)
- 2 programming assignment (10% each)
- 1 final examination (50%)

Plagiarism

- Any of the following constitutes an act of plagiarism
  - Submitting the work of another student as your own, or asking someone else to do your assignment
  - Direct copying
  - Studying someone else's work and then rewriting it as your own
  - Using a substantial part of other's work in your program
  - Providing your assignment as the source for any of the above actions

Plagiarism

- Penalties
  - First Attempt – warning letter and zero mark for the component concerned
  - Subsequent Attempt – disciplinary actions (a published reprimand, suspension of study for a period of time, fine, or expulsion from the University)

Where to find information and seek help?

- All the lecture/tutorial notes and assignment sheets can be downloaded from the course Moodle page
- Important information will be posted in the news and announcements
- If you have questions about course materials, please post your questions in the general discussion forum
- If you have questions about assignments, please post your questions in the discussion forum for that specific assignment
- You are always welcomed to talk to your lecturer and tutors about any problems you encountered in this course